

WASP-36b

R-0699

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_260119 · created 2026-07-27T21:30:31+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461059.7745 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.049
Transit depth (Rp/Rs)^2	2.77 +/- 1.63 [%]
Orbital Inclination (inc)	79.19 +/- 2.86
Airmass coefficient 1 (a1)	0.55 +/- 0.31
Airmass coefficient 2 (a2)	0.37 +/- 0.27
Scatter in the residuals of the lightcurve fit is	56.2 %
Best Comparison Star	#1 - [593, 48]
Optimal Aperture	2.41
Optimal Annulus	6.26
Transit Duration (day)	0.035 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.049 vs 0.1368 (deviation 0.62σ)
Duration fit vs published	0.035 d vs 0.0758 d (deviation 1.28σ)
Mid-time O–C	5.8 min
Depth SNR	3.4
Residual scatter	56.2 %

Light curve

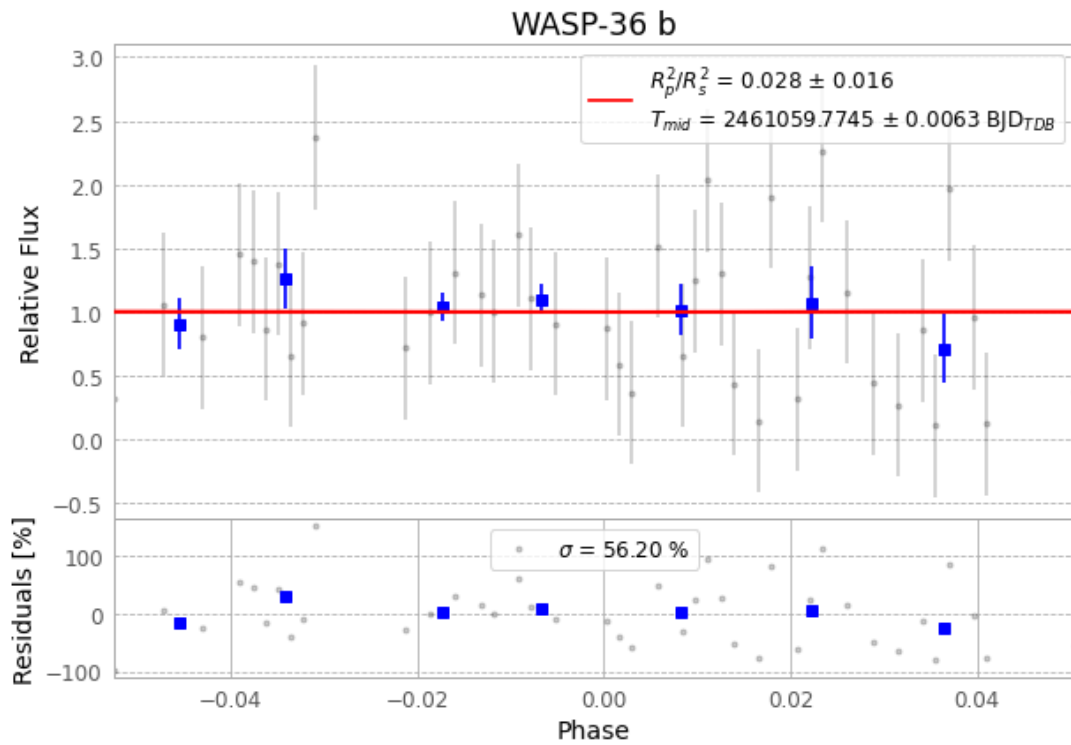


Figure 1 — FinalLightCurve_WASP-36 b_2026-01-18.png (SHA-256 755fa940dcb6f335...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [581, 393], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b34c4c0180b6d46c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

77 FITS frames (51 MB), WASP-36260119043015.FITS → WASP-36260119081815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-260119.log (da4f47790db7...), FinalLightCurve_WASP-36 b_2026-01-18.png (755fa940dcb6...), FinalLightCurve_WASP-36 b_2026-01-18.csv (4c8de52a8202...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 21:30Z.